

Proximate Deconfined Quantum Critical Point in $\text{SrCu}_2(\text{BO}_3)_2$

Y. Cui *et al.*, *Science* **380**, 1179 (2023)

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1 Introduction

- Deconfined Quantum Critical Point
- $\text{SrCu}_2(\text{BO}_3)_2$ and the Theoretical Model
- About this Paper
- Phase Diagram Overview

2 Experimental Method

- NMR spectrum
- Relaxation Rate

3 Discussion

- Weaken Discontinuity
- Numerical Simulation of J-Q Model
- Outlook

Introduction

Deconfined Quantum Critical Point

- **Ginzburg-Landau Paradigm**

The phase transition between two phases with unrelated symmetry breaking should be **first order**.

- **Deconfined Quantum Critical Point (DQCP)**

Beyond the GL theory. Phase transition with unrelated symmetry breaking could be **continuous**.

- **Example:** VBS-Néel transition

T. Senthil *et. al.*, *Science* **303**, 1490 (2004)

Deconfined Quantum Critical Point

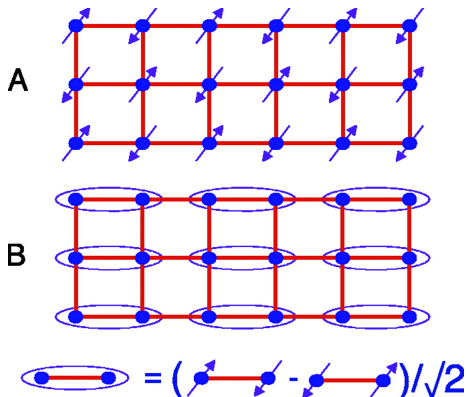


Figure: The VBS order breaks lattice translational symmetry while the Néel order breaks the $O(3)$ rotational symmetry. After T. Senthil *et. al.*, *Science* **303**, 1490 (2004)

Deconfined Quantum Critical Point

$$\mathcal{H} = J \sum_{\langle \mathbf{r}\mathbf{r}' \rangle} \hat{\mathbf{S}}_{\mathbf{r}} \cdot \hat{\mathbf{S}}_{\mathbf{r}'} + \dots$$

$$\hat{\mathbf{n}} = z^\dagger \boldsymbol{\sigma} z$$

$$\mathcal{L} = \sum_{\alpha=1}^2 |(\partial_\mu - i a_\mu) z_\alpha|^2 + s |z|^2 + u |z|^4 + \kappa (\epsilon_{\mu\nu\kappa} \partial_\nu a_\kappa)^2$$

T. Senthil et. al., *Science* **303**, 1490 (2004)

$\text{SrCu}_2(\text{BO}_3)_2$ and the Theoretical Model

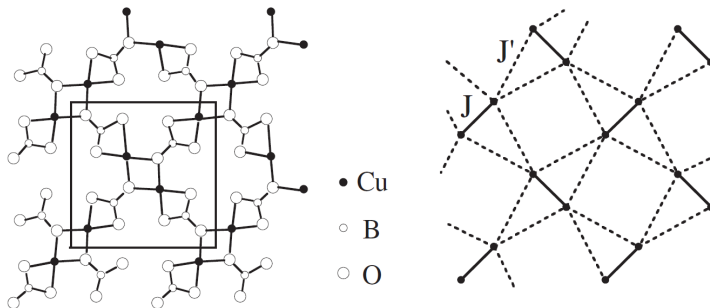


Figure: Structure of the magnetic layer in $\text{SrCu}_2(\text{BO}_3)_2$.

K. Kodama *et. al.*, *J. Phys.: Condens. Matter* **14**, L319 (2002)

$\text{SrCu}_2(\text{BO}_3)_2$ and the Theoretical Model

- **Shastry-Sutherland Model (SSM)**

$$\mathcal{H} = J \sum_{\langle ij \rangle} \hat{\mathbf{S}}_i \cdot \hat{\mathbf{S}}_j + J' \sum_{\langle ij \rangle \in \square_s} \hat{\mathbf{S}}_i \cdot \hat{\mathbf{S}}_j - h \sum_i \hat{S}_i^z$$

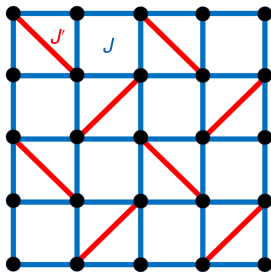


Figure: Shastry-Sutherland Model (SSM) on a square lattice. *Nat. Phys.* **15**, 678 (2019)

$\text{SrCu}_2(\text{BO}_3)_2$ and the Theoretical Model

$$\mathcal{H} = -J \sum_{\langle ij \rangle} P_{ij} - Q \sum_{ijkl \in \square_s} (P_{ij}P_{kl} + P_{ik}P_{jl}) - h \sum_i S_i^z$$

$$P_{ij} = \frac{1}{4} - \hat{\mathbf{S}}_i \cdot \hat{\mathbf{S}}_j$$

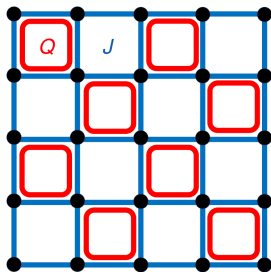


Figure: Checkerboard J-Q model. the J' term is transformed effectively to the Q term (4 spin coupling) here

$\text{SrCu}_2(\text{BO}_3)_2$ and the Theoretical Model

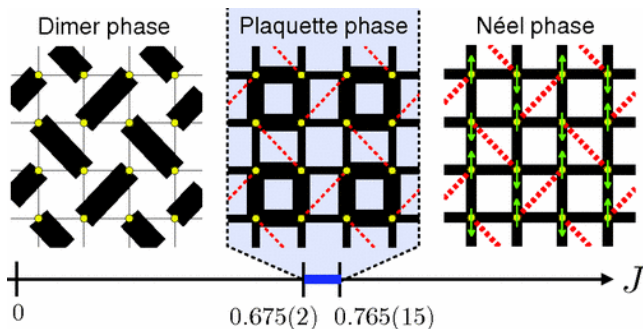


Figure: Three phases in the Shastry-Sutherland model (SSM) predicted numerically. *Phys. Rev. B* **87**, 115144 (2013)

About this Paper

- To confirm the phase diagram of $\text{SrCu}_2(\text{BO}_3)_2$ materials.
- Try to find the DQCP of the material $\text{SrCu}_2(\text{BO}_3)_2$ through nuclear magnetic resonance (NMR) method.
- Although they didn't find the DQCP actually, they pointed out the possibility of the existence of the DQCP and the possible future work to find it.

Phase Diagram Overview

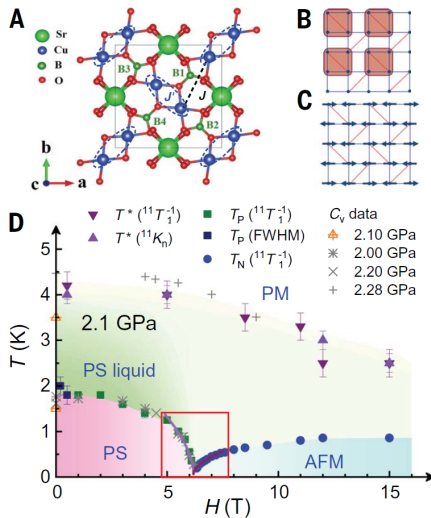


Figure: The phase diagram of $\text{SrCu}_2(\text{BO}_3)_2$ under pressure 2.1 GPa

Phase Diagram Overview

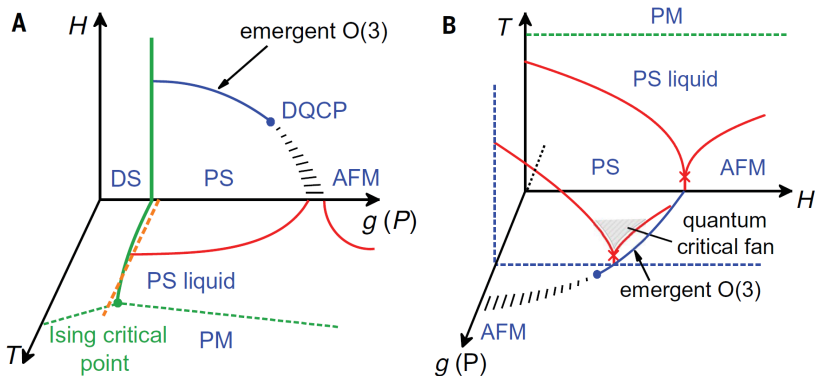


Figure: There are three parameters can be tuned. They are (i) **temperature**, (ii) external **magnetic field**, and (iii) **pressure**. Where the pressure can change the value of J'/J in the SSM.

Experimental Method

NMR Spectrum

The authors performed a ^{11}B NMR measurement

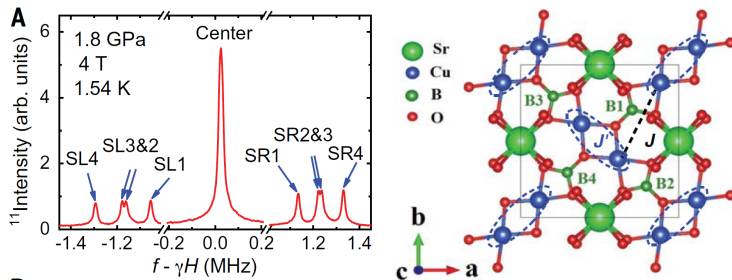


Figure: (Left) the ^{11}B NMR spectrum, including a central line and few satellite peaks arises from different positions of boron ions (right)

NMR Spectrum

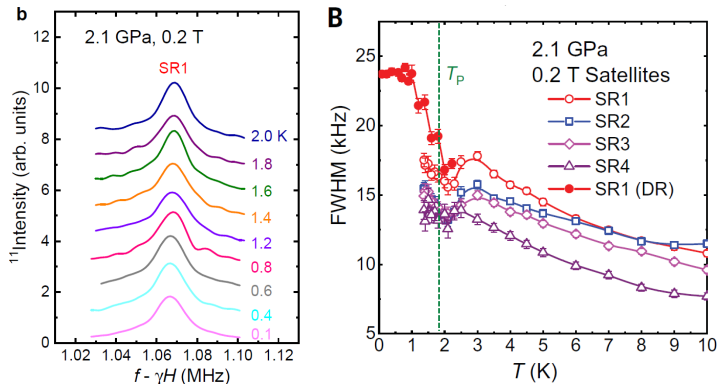


Figure: Full width at half maximum (FWHM) of each satellite peak at different temperature. The sharp inclination indicates the **formation of plaquette singlet (PS)** phase.

NMR Spectrum

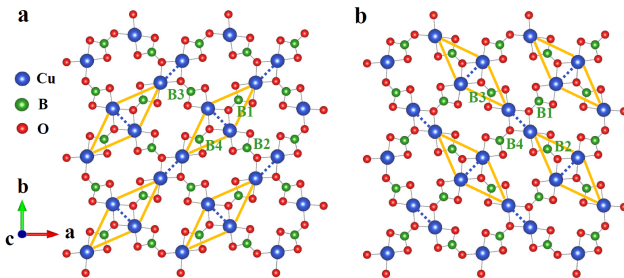


Figure: PS order formation leads to the lattice distortion, which broaden the satellite line width.

NMR Spectrum

Applying an **external magnetic field** (parallel to c -axis) induces central line splitting. Three peaks indicating the PS & AFM phase coexistence.

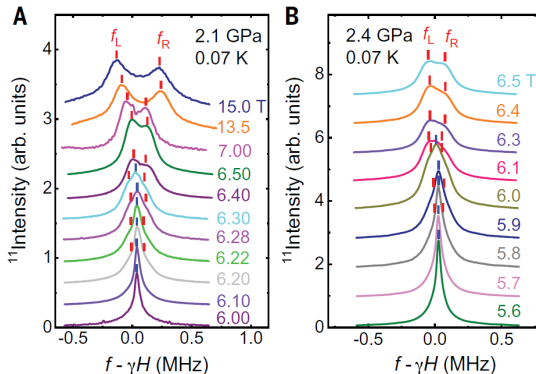


Figure: Field induced line splitting under pressure 2.1 GPa and 2.4 GPa respectively.

NMR Spectrum

(a) [100]			
site	H_{in}^x	H_{in}^y	H_{in}^z
B1	-0.011	-0.608	-0.168
B2	0.011	-0.608	0.168
B3	0.011	-0.608	-0.168
B4	-0.011	-0.608	0.168

(b) [010]			
site	H_{in}^x	H_{in}^y	H_{in}^z
B1	-0.608	-0.011	-0.168
B2	-0.608	0.011	-0.168
B3	-0.608	0.011	0.168
B4	-0.608	-0.011	0.168

(c) [110]			
site	H_{in}^x	H_{in}^y	H_{in}^z
B1	-0.619	-0.619	-0.336
B2	-0.597	-0.597	0
B3	-0.597	-0.597	0
B4	-0.619	-0.619	0.336

Figure: The **hyperfine field** at each boron ion sites.

The **spin-lattice relaxation rate** is a direct indicator of the low energy **spin fluctuation** (hence related to the magnetic susceptibility).

$$\frac{1}{T_1} \propto \frac{T}{\mu_B^2} \lim_{\omega \rightarrow 0} \sum_{\mathbf{q}} F_{\alpha}(\mathbf{q}) \frac{\chi''(\mathbf{q}, \omega)}{\omega}$$

$1/T_1$ and χ (spin fluctuation) are related through the **spin hyperfine interaction**

$1/T_1$ is a good indicator to clearly show the **PS-AFM** phase transition.

Relaxation Rate

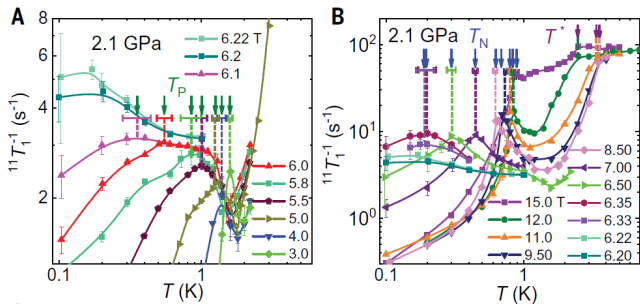


Figure: Temperature induced phase transitions under different magnetic field. (Left) A PS-PS liquid transition. (Right) An AFM-PS liquid transition.

Discussion

Weaken Discontinuity

Under a higher pressure, the discontinuity in $(f_R - f_L)$ is weakened, indicating a weak first-order transition.

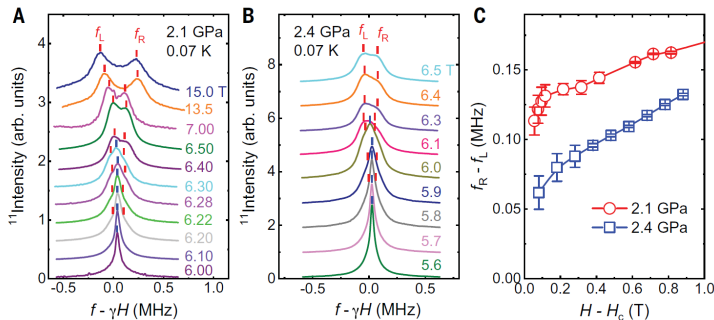


Figure: Field induced line splitting. The discontinuity in $(f_R - f_L)$ shows that the phase transition is first order.

Weaken Discontinuity

Scaling behaviour appears under a higher pressure (2.4 GPa)

$$\chi \propto 1/T_1 \sim aT^\eta$$

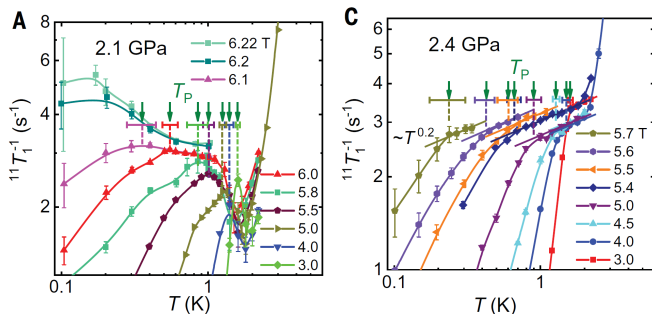


Figure: temperature induced phase transition with $1/T_1$ as the indicator.

Numerical Simulation of J-Q Model

Another important concept about DQCP is the **emergent symmetry**.

The authors applied quantum Monte Carlo method to showed that emergent symmetry of the **J-Q model in a magnetic field** is $O(3)$.

$$m_p = \frac{1}{L^2/2} \sum_{i \in \square_s} (-1)^{i_x} S^z(i) S^z(i + \hat{x}) S^z(i + \hat{y}) S^z(i + \hat{x} + \hat{y})$$

Emergent order parameter:

$$(m_x, m_y) \otimes m_p \rightarrow (m_x, m_y, m_p)$$

the combination becomes a $O(3)$ vector, corresponding to an $O(3)$ non-linear sigma model.

Numerical Simulation of J-Q Model

In the presence of $O(3)$ symmetry, the probability distribution of m_p should be uniform (like a flat plateau).

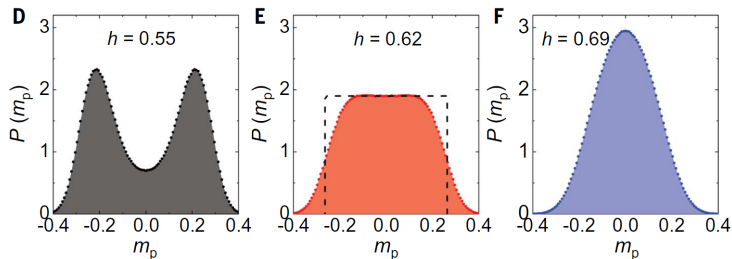


Figure: (Left) PS phase, Ising-like symmetry. (Middle) DQCP, emergent $O(3)$. (Right) AFM phase (no PS order, thus the average value is zero).

So the numerical result suggest an emergent $O(3)$ symmetry.

- To further increase the pressure to see if there indeed exist a DQCP.
- Identify the fractional excitations and the emerged gauge field.

THANKS FOR LISTENING

Knight Shift here indicates the changing of magnetic susceptibility χ .

$$K_{ii} = \mu_B N_A \chi A_{ii}$$

where **A** is the hyperfine interaction tensor

$$H_i = A_{ij} m_j$$

K. Kodama *et. al.*, *J. Phys.: Condens. Matter* **14**, L319 (2002)

Supplement-Knight Shift

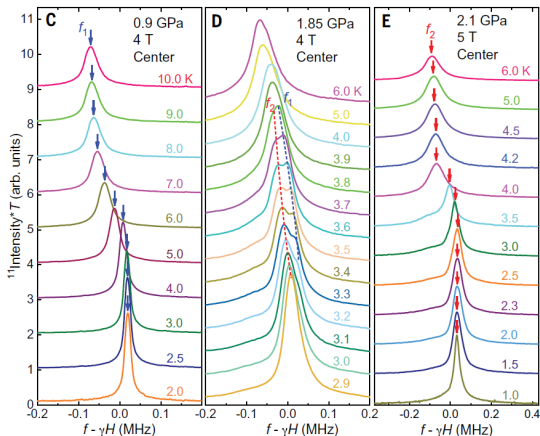


Figure: (Left) Central peaks in the DS phase. (Middle) DS and PS phase coexistence. (Right) Central peaks in the PS phase. The rapid change of the Knight shift in the right panel indicates a PS-PS liquid transition.

Supplement-Other Numerical Simulation

Another research indicated (using quantum Monte Carlo method) the emergent enhanced symmetry is $O(4)$ by plotting the probability $P(m_z, m_p)$ where m_p is the PS order parameter.

Notice that the system here is not under a magnetic field. The magnetic field, as shown in the main paper, breaks the original $O(3)$ Néel order parameter to $O(2)$, therefore the final dimension enlarged order parameter is (m_x, m_y, m_p) , which possesses an emergent $O(3)$ symmetry.

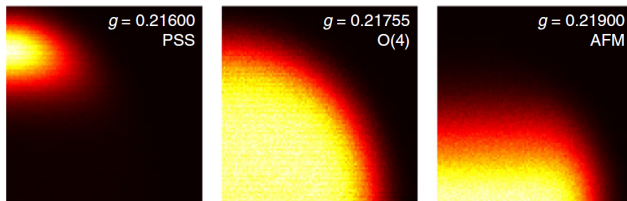


Figure: A numerical result confirmed the emergent $O(4)$ symmetry. *Nat. Phys.* **15**, 678 (2019)

The spin-lattice relaxation rate possesses the following relation with temperature

$$\frac{1}{T_1} \propto T^\alpha e^{-\Delta/k_B T}$$

where Δ is the spin gap.

There will be a sharp drop of the **relaxation rate** (which is actually proportional to the **spin fluctuation**, hence be proportional to the **magnetic susceptibility** χ) because of the existence of the gap.